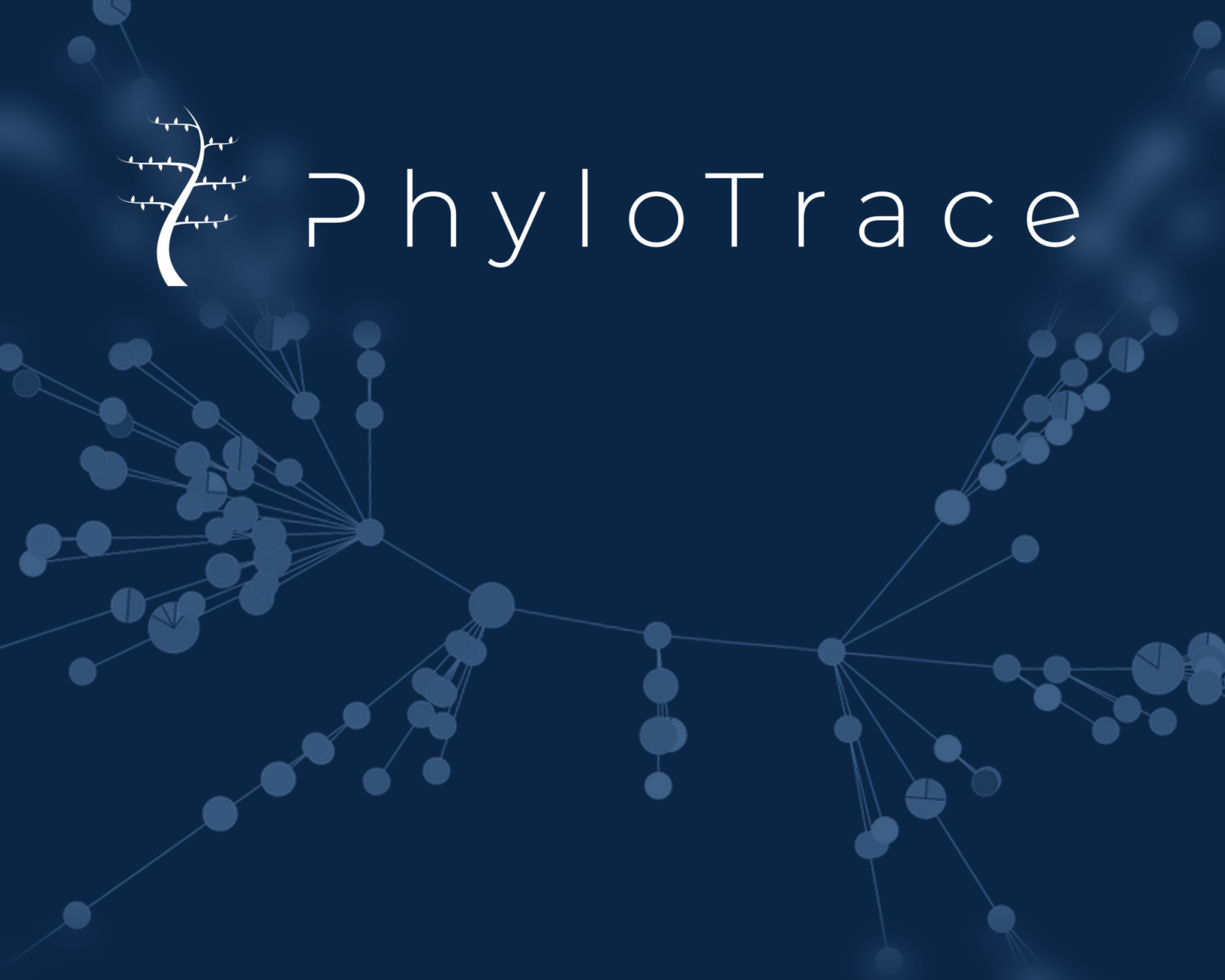
PhyloTrace



Decentralised pathogen surveillance — under your control.

Connect genomic information with epidemiological data to monitor, investigate and look back — in healthcare, food and environment.

Open source
Free software under GPLv3

No coding
One graphical, all-in-one platform

Runs locally
You decide what leaves your lab

Comparable
Across labs, without a central server

For research use only.

LIORA BIOINFORMATICS

THE CHALLENGE

Sequencing has scaled. Analysis hasn't.

Whole-genome sequencing shows how pathogens spread — through hospitals, food chains and the environment — often long before observation alone would. Many laboratories now sequence routinely. Far fewer can interpret those genomes themselves, together with the epidemiological data that gives them meaning.

7.7 M
deaths associated with 33 bacterial pathogens in 2019 — 13.6 % of all deaths worldwide[1]

1.27 M
deaths attributable to bacterial antimicrobial resistance in 2019[2]

600 M
people — almost 1 in 10 — fall ill each year after eating contaminated food[3]

Genomes leave the lab to be analysed.

Results return late, while an investigation loses value by the day — and epidemiological judgement concentrates in a few institutions.

Comparability needs coordination.

Locally assigned allele numbers drift apart between laboratories unless they are synchronised with a central authority, so every participant depends on it.

Sharing becomes a governance problem.

Pooling data in one place means sensitive metadata — about patients, producers or sites — has to move too.

> **Experts call for analysis capacity in the laboratories that collect samples.[4] That takes software they can run themselves.**

WHAT LABORATORIES NEED

Analysis on site
Turn genomes and epidemiological data into answers on site, close to the people who know the situation.

Comparable results
Results that line up across laboratories, without a central authority.

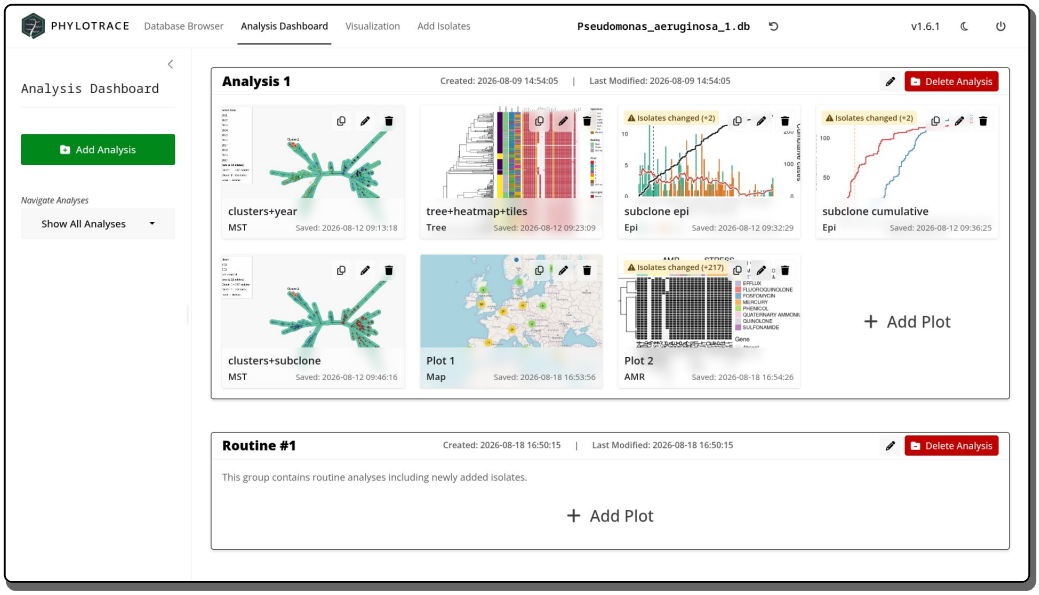
Control over data
Share what a collaboration needs — and nothing more.

[1] Ikuta et al., The Lancet, 2022, doi:10.1016/S0140-6736(22)02185-7 · [2] Murray et al., The Lancet, 2022, doi:10.1016/S0140-6736(21)02724-0 · [3] WHO, Estimates of the global burden of foodborne diseases, 2015 (31 hazards: bacteria, viruses, parasites, toxins, chemicals) · [4] Baker, Jauneikaite, Wheeler et al., Lancet Microbe 4(12), 2023.

PHYLOTRACE AT A GLANCE

Genomes and epidemiology in one platform.

PhyloTrace is an open-source platform for pathogen surveillance and monitoring. It integrates genomic data with epidemiological metadata, so you can follow what is happening now and look back at what has already happened — in public health, healthcare, food-associated and environmental settings, on your own hardware and without writing code.



The analysis dashboard: saved analyses per collection, including routine groups that grow with newly added isolates.

1 Genomes and epidemiology together

Typing results, resistance findings and your own metadata — time, place, source, any variable — are analysed side by side in one place.

2 Ongoing and retrospective

Monitor continuously as isolates arrive, or analyse existing collections retrospectively at any time. Saved analyses reopen exactly as you left them.

3 Comparable without a central server

Every allele is identified by a fingerprint of its sequence, so partner laboratories get identical results without coordinating.

4 Verifiable and under your control

Every result keeps its provenance — assembly fingerprint, scheme version, thresholds and tool versions — so it stays explainable years later. Analysis runs on your own hardware, and nothing leaves your laboratory unless you share it.

USE CASES

From routine monitoring to looking back.

PhyloTrace works the same way whether you watch for what is emerging or investigate what has already happened — in public health, healthcare, food-associated and environmental settings.

Surveillance works on two levels: fast local results to recognise and contain outbreaks, and comparison across institutions and countries that supports the search for resistant clones or food-chain problems — in a One Health perspective.[5]

	Public health	Food-associated	Environmental
Ongoing monitoring Saved analyses reopen as new isolates arrive.	**Spot clusters early** Routine typing of ward isolates reveals unnoticed clusters before they become outbreaks.	**Watch for recurring strains** Monitor isolates from production sites and products to see whether the same strain keeps returning.	**Track reservoirs over time** Follow resistant strains in water, surfaces or drains to find reservoirs before they cause harm.
Retrospective analysis Bring in existing collections and explore them at any time.	**Re-examine an outbreak** Add isolates or metadata to a past outbreak and confirm or rule out transmission routes.	**Trace back a contamination** Compare historical product and patient isolates to find the likely source.	**Test a suspected reservoir** Check whether an environmental source, such as a sink drain, was linked to an earlier hospital cluster.

Across all settings: one platform, one comparable language.

Because alleles are identified by a fingerprint of their sequence, laboratories from public health, healthcare, food and environmental settings can compare results without pooling their data.

[5] Muloi et al., The Lancet Microbe 4(12), 2023; Struelens et al., Frontiers in Science 2, 2024.

INSIDE THE PLATFORM

Everything in one place.

PhyloTrace covers the whole surveillance routine in five areas and keeps everything together in a single portable file. Work wherever you need — start a new collection, keep an ongoing one up to date, or bring in existing data and look back.

Set up	Type & screen	Curate	Analyse	Share
One portable file				

Set up

Choose from 42 schemes covering the most relevant bacterial species, ready to use and with nothing to convert by hand. Support for custom and additional schemes will be added.

Type & screen

Add genomes one at a time or type large batches automatically, whole existing collections included. Each genome is typed and screened for resistance and virulence in a single run.

Curate

Add epidemiological information — wards, specimen or food sources, sampling sites — and any variable you define. Many quality indicators are shown for every isolate.

Analyse

Explore relatedness, time and place with trees, epidemic curves, maps and resistance heatmaps. Save analyses for routine monitoring, or revisit historical collections at any time.

Share

Export any subset of isolates and choose field by field which information travels with them. Partner laboratories get what a collaboration needs while sensitive metadata stays in yours.

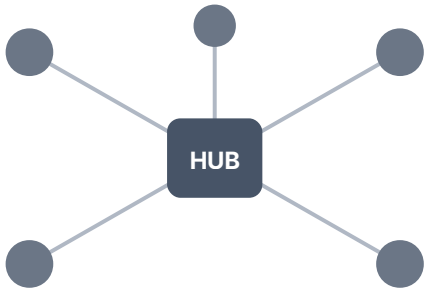
One file holds everything.

Genomes, metadata, results and saved analyses all live in one portable file — a standard SQL database that common tools can open, so your data is never locked in. To back up, copy it. To share, send it.

SURVEILLANCE NETWORKS

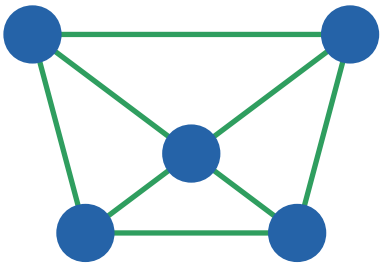
Collaborate without centralising.

Traditional surveillance pools every dataset at a central hub. By design, every participant then depends on that hub: whether it is reachable, how it is configured and what its rules and governance allow. Whatever the hub cannot provide, no partner can do alone. A decentralised network of laboratories — public health, healthcare, food and environmental — reduces that dependence: each laboratory keeps full analytical capability on its own hardware, decides what it shares and compares results directly with its partners.



Central hub

All data and analysis flow to one place — one shared dependency for every participant



PhyloTrace network

Labs analyse locally and exchange directly — each stays operational on its own

- **The same allele, the same name, everywhere**

Alleles are identified by a fingerprint of their sequence. Partners get identical identifiers without coordinating — no renumbering, no central nomenclature server to stay synchronised with.[1]

- **Exchange without extra steps**

Data harmonisation is built in: species and scheme are matched on import, a backup is made automatically and every isolate carries a label of its origin. Partner profiles appear directly in your trees and networks — no conversion, no in-between steps.

- **You decide what travels**

Export any subset of isolates and choose field by field which metadata goes with them — share relatedness while patient identifiers stay in your lab.

- **Data stays where it belongs**

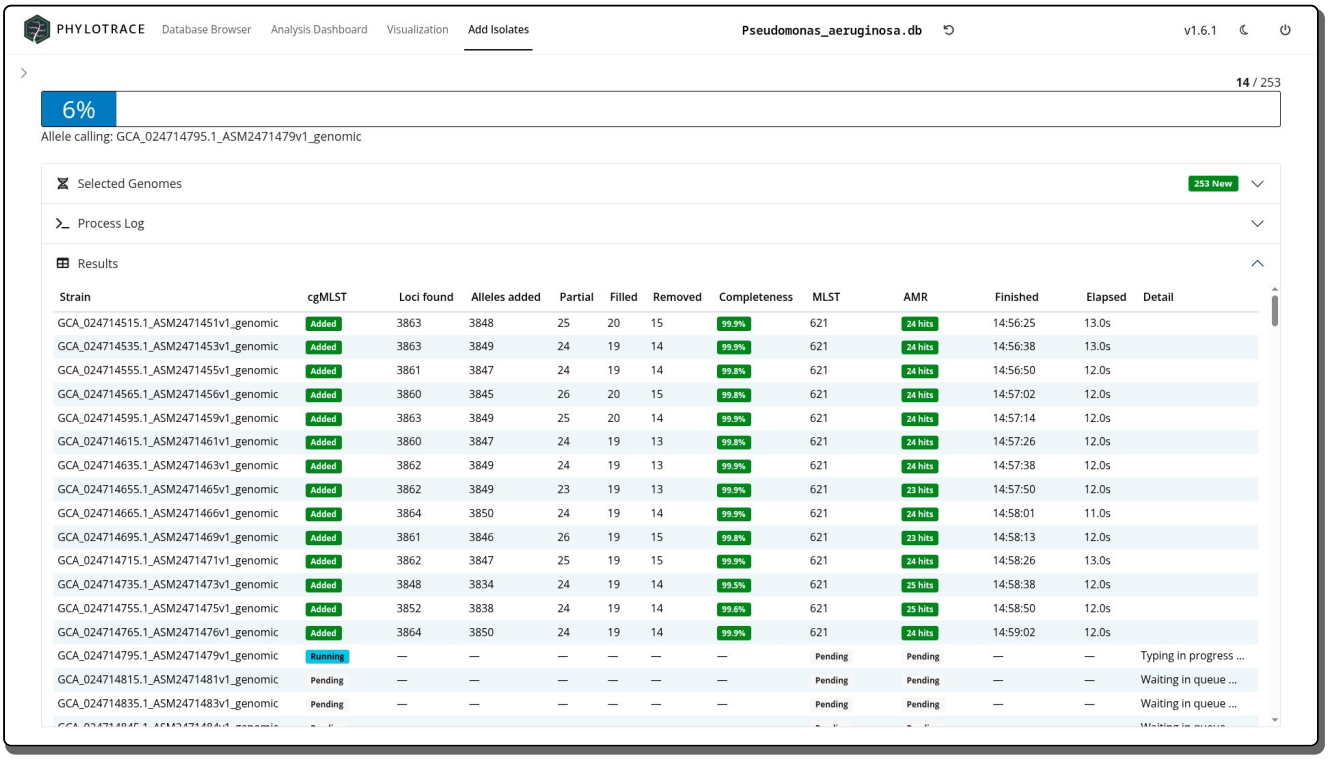
Confidential metadata stays with whoever is responsible for it. Each laboratory analyses its own samples on its own computer.

> **Each laboratory can keep analysing and comparing whenever a partner or a central service is unavailable — and stays in control of its data.**

[1] On the dependence on central nomenclature servers in outbreak settings: Deneke et al., Frontiers in Microbiology 12:649517, 2021, doi: 10.3389/fmicb.2021.649517.

FEATURES · TYPING & DATA

Typing, screening and curation in one place.



Typing run: cgMLST, MLST and resistance screening per assembly.

cgMLST and MLST typing

Fast and automated: 253 genomes in 49 minutes on a desktop workstation (page 9). cgMLST gives high resolution; classical MLST runs alongside for comparability with published sequence types. Only intact coding sequences become new alleles — **so every comparison starts from clean data and stays comparable over time.**

Resistance, virulence and point mutations

Every assembly is screened with NCBI AMRFinderPlus for resistance genes, virulence factors, stress, biocide and metal tolerance and species-specific resistance mutations — **so resistance is part of every analysis, not a separate project.**

Collection overviews and quality control

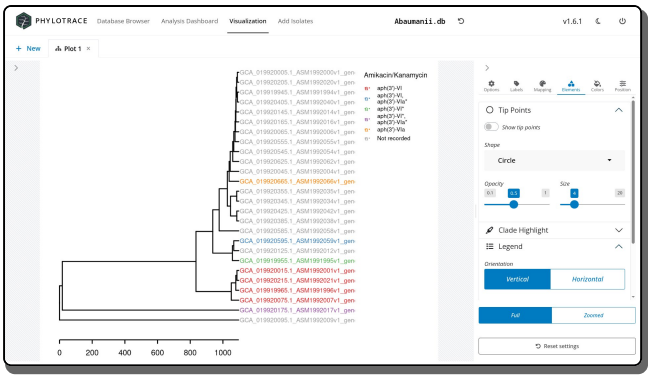
Dedicated overviews summarise provenance, assembly quality, profile completeness, sequence types and the settings and versions of every run. Consistency, completeness and duplicate checks flag weak data — **so you see exactly what your collection holds and weak data never distorts clusters.**

Your metadata, your variables

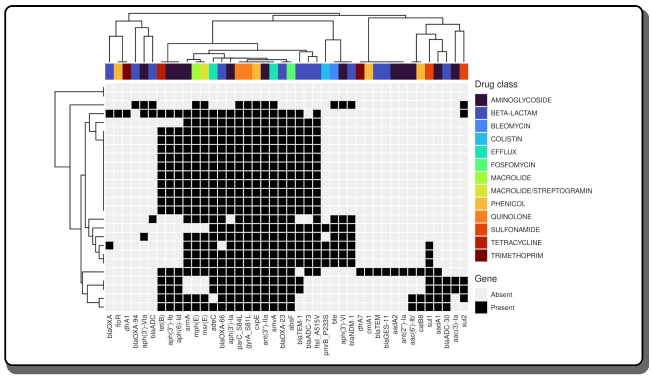
Standard fields such as date, source and location, plus custom variables of any kind — wards, environmental data, susceptibility results — **so relatedness is always read in its epidemiological context.**

FEATURES · ANALYSIS & VISUALISATION

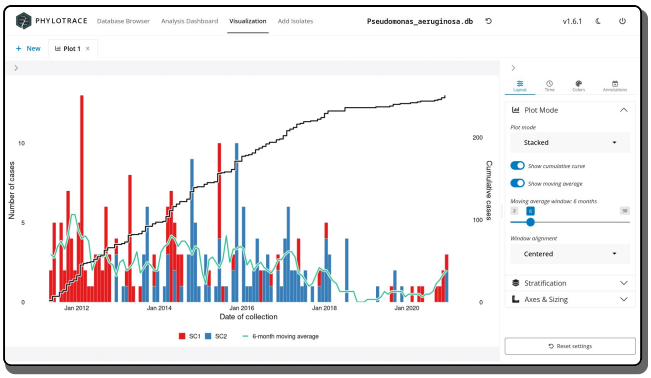
See what your genomes are telling you.



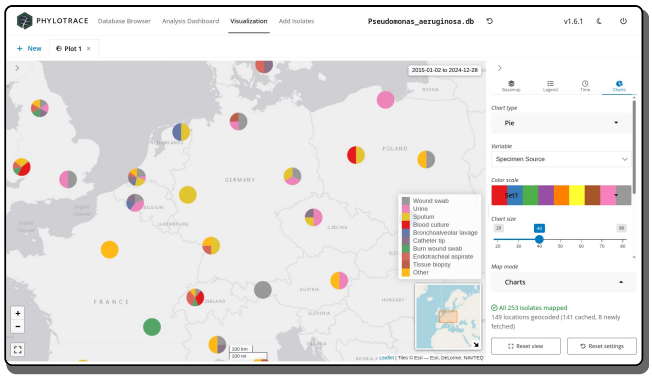
Phylogenetic tree with resistance panel



Resistance heatmap



Epidemic curve



Geographic map

Minimum spanning trees

Identical profiles grouped, every edge labelled with its allelic distance, clusters called at the scheme's published threshold, and any variable added as colour or label — **so you see which isolates belong together and uncover epidemiological population dynamics.**

Phylogenetic trees

Neighbour-joining or UPGMA, rectangular or circular, annotated with any variable — from wards and sources to resistance heatmaps — fully customisable, **so lineage, resistance and epidemiological context are read together.**

Epidemic curves and maps

Bin cases by day, week, month or year and play curve and map forward through time — **so you see when and where transmission happened.**

Resistance heatmaps

View resistance by gene, by drug class or as ranked prevalence — **so emerging resistance stands out across your collection.**

Smart variable mapping

Put any metadata field on any graph; PhyloTrace picks a fitting colour scheme and encoding — **so you explore hypotheses instead of formatting plots.**

Analysis dashboard

Saved analyses reopen exactly as you left them, flag when the underlying data has changed and export as PDF, SVG or PNG — **so routine monitoring is repeatable and takes only a few clicks.**

PROVEN ON REAL DATA · RETROSPECTIVE ANALYSIS

A decade-long outbreak, typed in 49 minutes.

We analysed a published, independently studied collection retrospectively: a *Pseudomonas aeruginosa* outbreak in a 425-bed US hospital, spread across 26 wards over ten years and sustained by reservoirs in sink drains.[1]

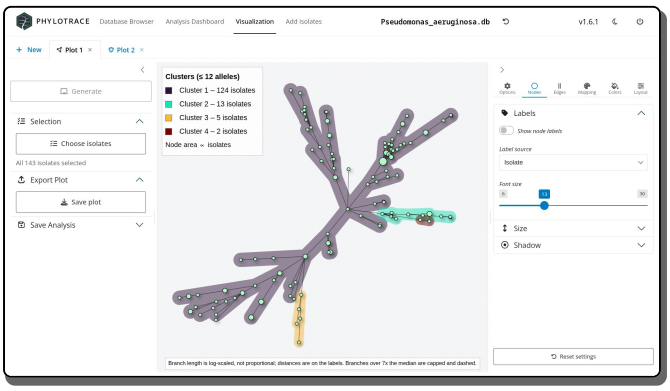
253
genomes typed

3,867
cgMLST loci per genome

ST-621
classical MLST, as published

4,176
resistance genes and mutations detected

- ✓ All 253 genomes typed and screened on a desktop workstation at default settings, with no manual tuning.
- ✓ Resolved into one outbreak cluster at the scheme's published threshold.
- ✓ Detected the two resistance determinants behind the outbreak's resistance evolution, with the published subclone distribution.
- ✓ Genomic relatedness and epidemiological metadata merged: epidemic curves by subclone and by building reveal how the outbreak developed over time.



Example view: minimum spanning tree in PhyloTrace.

Full details in the application note
liora-bioinformatics.com/phylotrace-application-note

[1] Stribling et al., eLife 13:RP93181, 2025; data from NCBI BioProject PRJNA852179. PhyloTrace manuscript in preparation.

HOW PHYLOTRACE COMPARES

The integrated experience of commercial software — open, decentralised and professionally supported.

	Command-line tools	Visualisation tools	Web platforms	Commercial desktop software	PhyloTrace
Graphical interface, no coding	–	✓	✓	✓	✓
Types genomes itself	✓	–	✓	✓	✓
Resistance screening built in	–	–	◐	✓	✓
Metadata and epi views in one tool	◐	◐	✓	✓	✓
Data stays in your lab	✓	✓	–	✓	✓
Comparable without a central server	◐	n/a	–	–	✓
Open source	✓	✓	◐	–	✓
No licence fee for the software	✓	✓	◐	–	✓
Professional support and onboarding	–	–	◐	✓	✓

✓ typically yes · ◐ partly · – typically no

Command-line tools are powerful and scalable, but mostly cover single steps and require command-line skills. Web platforms provide valuable public context, but require uploading your data to a service you don't operate. Commercial desktop software typically offers an integrated experience, but is licensed per user and term, and often relies on a central nomenclature server. PhyloTrace is free to use; you subscribe only for the support you need.

> **PhyloTrace combines integration, openness and independence — with professional support when you need it, as a flat subscription instead of a licence.**

Comparison by tool category, based on publicly available information; individual products may differ.

TRUST & SUSTAINABILITY

Open, local and built to last.

Surveillance software is a long-term decision. PhyloTrace is built so that your data stays yours and the software stays available.

YOUR DATA, YOUR JURISDICTION

Local by design

All analysis runs on your own hardware. Genomes and patient-related metadata leave your laboratory only if you share them.

Data protection

PhyloTrace is designed with data protection in mind: genomes and metadata stay on your own hardware. Liora hosts no customer data, has no permanent access to it and collects no telemetry.

Made in Germany

Developed and supported by Liora Bioinformatics in Rottweil, with research roots at Furtwangen University.

Research use only

PhyloTrace is not a medical device. Laboratories validate results for their own purposes under their own quality system.

Built on established methods

Hash-based allele identity is an established principle (Eyre 2019; Deneke 2021). PhyloTrace builds on pyMLST and AMRFinderPlus and works alongside public platforms.

OPEN SOURCE THAT LASTS

Yours to keep

Licensed under GPLv3 — the source code stays available to you, whatever happens.

Public and archived

Code on GitHub, archived on Zenodo (DOI 10.5281/zenodo.10996423).

Releases & roadmap

Regular maintenance updates and new features. Planned next: plasmid typing, support for cgMLST schemes beyond those currently included, and assembly generation — taking PhyloTrace end to end, from sequencing data to surveillance result.[1]

Maintenance you can rely on

Open source brings transparency and independence; its risk is that maintenance depends on goodwill. Subscriptions fund it: for subscribers, support for the current and the previous public version.

Academic backing

Developed with Furtwangen University (HFU) and the Medical University of Graz (MUG); peer-reviewed publication in preparation.

[1] Roadmap subject to change.



SERVICES & SUPPORT BY LIORA

Free software. Professional support.

PhyloTrace is and will remain open source. Open source brings transparency, independence and no lock-in — and it can leave a laboratory alone with installation, updates and open questions. Without active maintenance, open-source software falls behind: dependencies change, tools stop working and fixes no longer arrive. Our subscription mitigates those risks and keeps the benefits: someone who keeps PhyloTrace running, answers your questions and grows it with your needs.

SUBSCRIPTION

Support & maintenance

A direct line to the developers, priority and earlier access to updates and new features, and maintained current and previous public versions — so problems are solved by the people who build PhyloTrace.

SUBSCRIPTION

Onboarding & training

Installation on your infrastructure and hands-on training for your team, so routine surveillance runs without writing code.

PROJECTS

Custom development

Additional analyses, custom feature requests, integrations into your laboratory systems and custom analysis solutions — built by the team behind PhyloTrace. On request, this includes building infrastructure for networks of cooperating laboratories.

WHY SUBSCRIBE IF THE SOFTWARE IS FREE?

Reliable operation

Problems are handled for you, with a direct line to the developers and priority access to updates.

Faster routine

Onboarding and training get your team productive without building in-house bioinformatics.

Sustained development

Subscriptions fund the maintenance and further development of PhyloTrace as open source.

Compare plans → page 13 · Get started → page 14

PLANS

Free to use.
Supported when you need it.

PhyloTrace itself is free and open source. Plans add the support, services and commitments a routine laboratory needs.

	Community	Subscription
Price	**Free**	**€149 per month** or **€1,499 per year**[1] Minimum term 12 months
PhyloTrace software	✓	✓
Updates & new features	Public releases	Priority and earlier access
Public documentation & community support (GitHub)	✓	✓
Professional documentation	–	✓
Direct support from the developers	–	Target first response: 1 business day for critical and high priority, 2–5 business days otherwise
Installation & onboarding	–	✓
Training for your team	–	Remote onboarding session included; further training on request

Add-ons for every plan: custom development, feature requests and integration into laboratory systems, on request also infrastructure for networks of cooperating laboratories.

Not sure which plan fits? Get in touch: info@liora-bioinformatics.com

[1] All prices are net prices in euros, plus statutory VAT.

GETTING STARTED

From first look to routine use.

Four steps, at your pace. You can stop at any of them — the software stays yours.

1

Download

Install PhyloTrace for free from GitHub on a Linux workstation and try it with your own genomes.

2

Demo

We show you PhyloTrace with your own questions in mind — for your laboratory or your network.

3

Pilot

After the demo, run PhyloTrace on your own data for six months at a reduced fee.

4

Subscription

Choose the plan that fits your laboratory and move into routine use.

What you need to start

✓ **A desktop workstation**

Linux today; Windows support is planned. No server or cloud account required.

✓ **Genome assemblies**

Bacterial assemblies in FASTA format — from your own sequencing or a service provider.

Request a demo: info@liora-bioinformatics.com

After the demo: pilot programme

Duration

6 months of pilot operation, starting with go-live in week 2 after kick-off

Fee

€99 per month[1] (€594 for 6 months)

Everything included, step by step: the pilot offer

liora-bioinformatics.com/phylotrace-pilot · open to a limited number of laboratories

[1] All prices are net prices in euros, plus statutory VAT.

ABOUT LIORA BIOINFORMATICS · MADE IN GERMANY

Technical details, at a glance.

Liora Bioinformatics develops and maintains open-source software for life-science research. PhyloTrace was developed with Furtwangen University (HFU) and the Medical University of Graz (MUG) and presented at ShinyConf 2024 and R/Medicine 2024. Our aim is to lower the barrier to genomic surveillance for laboratories that cannot afford or accept existing solutions.

Runs on	Linux; Windows support planned. Desktop workstation (tested: 8 cores, 24 GB RAM).
Interface	Local application that opens in your web browser. All computation and data stay on your machine.
Installation	One installer sets up PhyloTrace and all the analysis tools it uses.
Input	Genome assemblies (FASTA). Integrated genome assembly is planned.
Data	One portable database file per project.
Allele identity	SHA-256 hash of each allele sequence; partner databases merge directly, and partner profile tables can be linked.
Exchange formats	Database file, Excel, CSV/TSV, FASTA
Figures	PDF, SVG, PNG
Licence	GNU GPL v3 · archived on Zenodo, DOI 10.5281/zenodo.10996423

For research use only. PhyloTrace is not a medical device and is not intended for use in diagnostic procedures, for guiding treatment decisions, or for the management of individual patients.

PhyloTrace

Ready to bring pathogen surveillance into your lab?

See PhyloTrace with your own questions in mind. We'll show you how it fits your laboratory or network.

REQUEST A DEMO

info@liora-bioinformatics.com

liora-bioinformatics.com/phylotrace

Website

GitHub

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78628 Rottweil
Germany

LIORA BIOINFORMATICS